Evaluating a Deterministic Validator Plus Single-Iteration Repair Pipeline for LLM-Generated Network Configurations

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***Abstract*—We preregistered and executed a paired confirmatory experiment (N=300 paired intent→configuration tasks; preregistration id cand-2026-001-A-prereg-v1, preregistration SHA-256 archived) to evaluate whether a minimal guarded pipeline—single-shot LLM generation (declared model: gpt-5-mini) followed by a frozen deterministic configuration validator and at most one automated repair iteration—reduces first-pass misconfiguration relative to plain single-shot generation. Execution completed 600 episodes and used 301 model calls. Observed outcomes: baseline = 299/300 valid (99.667%); guarded = 300/300 valid (100.0%); paired counts n11=299, n10=1, n01=0, n00=0. Exact McNemar two-sided p = 1.00; absolute paired difference = 0.0033333; paired bootstrap 95% CI = [0.00, 0.01] (10,000 resamples, seed 1729). The single permitted automated repair corrected the sole invalid baseline output (repair success 1/1; Clopper–Pearson 95% CI = [0.025, 1.0]). The preregistered planning assumption used for power calculations (planned baseline pass rate = 0.40) was contradicted by the observed near-ceiling baseline, removing statistical headroom for the preregistered >=50% relative reduction target. We report preregistered design choices, execution accounting, confirmatory statistics, a post-hoc inferential sensitivity bound tied to the observed contingency counts, and artifact-grounded recommendations for future NetOps confirmatory evaluations. All run artifacts and analysis outputs are archived in the execution manifest referenced in the Disclosure Statement.**

I. Introduction

Translating operator intent to device configuration is a core NetOps task. Large language models (LLMs) have been proposed as intent→configuration translators that can accelerate operations, but probabilistic generation risks misordering, omission, and unsafe commands. Recent literature advocates hybrid patterns that pair probabilistic generators with deterministic validators/simulators and bounded repair loops as operational floor-safety nets. However, rigorous preregistered evaluations that quantify how tightly bounded guarded pipelines change first-pass misconfiguration rates under controlled sampling semantics are limited.

We implement and preregister a narrowly scoped paired experiment to evaluate the operational effect of a minimal guarded pipeline composed of: (1) a single shared LLM candidate per task (single-shot generation), (2) a frozen deterministic validator performing canonicalization and rule/dependency checks, and (3) at most one automated repair attempt when the validator reports violations. The core research question is: Can this minimal guarded pipeline reduce first-pass misconfiguration rate relative to plain single-shot generation under a controlled paired design? Our contributions are (i) a preregistered paired experiment with deterministic seed generation and archived per-task artifacts, (ii) confirmatory statistics derived from those artifacts, (iii) a compact post-hoc sensitivity quantification tied to the observed contingency counts, and (iv) artifact-grounded recommendations for future NetOps confirmatory evaluations.

II. Related Work

Recent work on LLMs for NetOps and intent-driven configuration has produced numerous system demonstrations and surveys documenting both promise and concrete failure modes. Empirical demonstrations of intent→configuration translation report that LLMs can synthesize plausible device configuration templates in lab and testbed settings, yet they frequently surface operationally relevant errors such as command misordering, omission of required safety steps, and overconfident assertions that lack mechanical verification (example system evaluation and discussion: Nguyen et al. 2024 [10.1002/nem.2313]). Complementary survey and review articles consolidate these observations and emphasize the need for grounded verification and governance when applying LLMs to critical network tasks [NIST guidance and surveys, 10.6028/nist.ai.100-2e2023].

Parallel technical threads motivate guarded architectures. Work in adjacent domains (clinical QA, code repair, and general RAG/verification research) shows measurable reductions in factual errors when probabilistic generators are paired with retrieval, deterministic checks, or repair loops; these findings have been ported by several NetOps proposals arguing for generate→verify→repair pipelines rather than blind single-shot deployment [e.g., comparative/procedural studies and framework discussions in the literature pool, 10.36227/techrxiv.176784505.56675782/v1; 10.2139/ssrn.6608518]. Architectural proposals for intent-aware automated remediation and multi-agent NetOps orchestration similarly argue that deterministic validators and bounded repair/backout policies are central to safe agentic NetOps workflows (see recent agentic NetOps proposals and intent-translation demonstrations, e.g., 10.36227/techrxiv.177004277.71795055/v1).

Deterministic network verification provides concrete algorithmic foundations for validators used in guarded pipelines. Prior work on specification-based and header-space analyses demonstrates tractable static checks for reachability, ac-

cess control, and update-safety properties without requiring live device execution; such algorithms can be integrated to canonicalize generated artifacts into normalized ASTs and to flag violations that trigger repair logic [Beckett et al., 2017 foundation for incremental/specification checks, 10.1145/3098822.3098834]. These deterministic checks differ fundamentally from empirical or emulation tests: they provide provable, rule-based pass/fail decisions for a defined specification, but they do not by themselves measure runtime control-plane or packet-level effects unless combined with higher-fidelity simulation or emulation.

Limitations in the prior literature motivate the present work. Many demonstrations and proposals are system-level or exploratory rather than preregistered confirmatory experiments; reported improvements from RAG, verifier-assisted generation, or agentic architectures often lack paired within-task statistical controls that isolate the incremental effect of a validator+repair floor on the same generated candidate. Moreover, published comparisons rarely quantify how sampling semantics (shared candidate versus independent draws) and repair budgets affect the number of informative discordant pairs available to paired binary tests. Surveys and architectural reviews therefore call for reproducible evaluation harnesses that combine deterministic validators with controlled sampling semantics and explicit inferential plans [10.6028/nist.ai.100-2e2023].

Our study addresses a tightly scoped portion of this gap by executing a preregistered paired experiment that (a) fixes shared_initial_candidate semantics to measure remediation capacity on a single generated plan, (b) uses a frozen deterministic validator to define a binary success criterion tied to intent constraints, and (c) limits repair budget to a single automated iteration. This design directly tests the operational claim that a minimal validator+repair floor reduces first-pass misconfiguration, and it surfaces the inferential consequences of high baseline pass rates and shared-candidate pairing for confirmatory McNemar-style testing.

III. Methodology

Preregistration and hypotheses: The experiment was preregistered (cand-2026-001-A-prereg-v1) prior to observing outcomes. Confirmatory hypotheses were: H1 — the guarded pipeline reduces first-pass misconfiguration rate by >=50% (relative) versus single-shot generation; H2 — one or fewer automated repair iterations recover >=75% of initially invalid configurations. The preregistration explicitly used a planning assumption of planned baseline pass rate = 0.40 for power calculations; this numeric assumption is central to the interpretation of the confirmatory outcome and is documented in the preregistration record (see Disclosure).

Design and sampling semantics: We adopted a paired within-task design with 300 deterministically generated seeds from a frozen task generator. For each seed the same sampled LLM candidate was reused in both arms (shared_initial_candidate semantics): baseline (single-shot generation) and guarded (identical candidate validated and—if invalid—subjected to up to one automated repair attempt). Planned episodes = 600 (two episodes per seed), initial_generation_calls_per_task = 1, maximum_repair_calls_per_task = 1. The shared-candidate pairing was preregistered to isolate the incremental effect of validation and repair on a fixed generated plan; this choice increases within-pair correlation and reduces the number of informative discordant pairs available to McNemar inference compared with an independent-draws design.

Model, validator, and scoring: The declared model in the execution contract was gpt-5-mini. The frozen deterministic validator/repair adapter (validator id deterministic_netops_validator_v5) implements rule and dependency checks, canonicalizes configurations to normalized ASTs, and returns binary pass/fail scoring plus violation codes used to trigger and steer the preregistered automated repair adapter. The scoring rule is full intent-constraint validation: outputs are scored as success only when required command sequences and workflow constraints are satisfied by the validator's canonicalization and rule checks. The preregistered missingness plan allowed one retry for provider failures; primary analysis used the 'complete_pair_primary' treatment which drops pairs only if both calls fail. All validator decisions, normalized configurations, and repair traces were archived in the execution manifest and analysis bundle.

Power, estimand, and analysis plan: The primary estimand was the paired_success_rate_difference_guarded_minus_baseline. The preregistered primary test was the exact McNemar two-sided test (α=0.05) on paired pass/fail outcomes; we also preregistered paired bootstrap percentile 95% CIs (10,000 resamples) for the absolute paired difference. The preregistration included a planning assumption baseline pass rate = 0.40 which informed the sample size and the >=50% relative reduction target; this assumption is reported here to make the planning calculus transparent. Secondary analyses included exact binomial intervals for repair success, stratified checks, and sensitivity analyses treating failed model calls as failures; all analysis code and per-task artifacts are archived.

IV. Results

Execution accounting and artifact provenance (archived): The preregistered run completed all planned episodes: planned episodes = 600, completed episodes = 600, complete analyzed pairs = 300. Model calls consumed = 301 total (300 shared initial generation calls + 1 repair invocation). The archived deterministic reconciliation/provider audit records show no provider or infrastructure failures and no deterministic exclusions for contamination under the preregistered checks. These execution counts and stage summaries are recorded in the archived execution manifest and analysis bundle and are the direct source for all numeric values reported below.

Primary per-condition outcomes (archived counts): Baseline (single-shot) success count = 299/300 (99.6667%); Guarded (validator + <=1 repair) success count = 300/300 (100.0%).

The paired contingency table derived from the archived analysis artifact is: n11 (both success) = 299; n10 (guarded success, baseline fail) = 1; n01 (guarded fail, baseline success) = 0; n00 (both fail) = 0. Missingness summary in the archive reports zero failed or unscorable episodes in either arm under the preregistered missingness treatment.

Confirmatory inferential results (preregistered analyses executed on archived artifacts): The exact McNemar two-sided test on the observed contingency yields p = 1.00. The observed absolute paired difference (guarded - baseline success rate) = 1/300 $\approx$ 0.0033333. Paired bootstrap percentile 95% CI for the absolute paired difference = [0.00, 0.01] (10,000 resamples, bootstrap seed = 1729), as computed in the archived analysis artifact. Repair outcomes: the single preregistered automated repair invocation succeeded on the sole initially invalid baseline output (repair success 1/1). The exact Clopper–Pearson 95% CI for a 1/1 repair success is [0.025, 1.0], reflecting the large uncertainty from a single observed repair case.

Why the McNemar p = 1.00 and what the discordant count implies: Exact McNemar inference is driven entirely by the discordant pair counts (n10 and n01). With only one discordant pair observed (n10+n01 = 1), the two-sided exact test cannot reject the null of no difference; the minimal nonzero discordant total yields an exact two-sided p of 1.00 under the test implementation used in the preregistered analysis. Equivalently, the discordant total places a hard upper bound on the resolvable absolute paired difference equal to discordants / 300 = 1/300 $\approx$ 0.00333. Put practically, the observed near-ceiling baseline left essentially no statistical headroom to detect the preregistered >=50% relative reduction (which, given the preregistration's planning baseline = 0.40, mapped to an absolute change on the order of +0.30). Detecting an absolute +0.30 with N=300 would require on the order of tens of discordant pairs (the preregistration noted ~90 net discordant pairs aligned with the planned effect size under conservative assumptions); the archived discordant count of one is far below that requirement. These limits and numeric bounds are computed from the archived contingency artifact and bootstrap outputs.

Representative validator and diagnostic traces (artifact-grounded): For each episode the archived validator trace contains structured diagnostics used for scoring: normalized_configuration, parsed_command_count, required_command_count, observed_touched_field_count, violation_codes, validation_before/after states, and a repair_applied flag. In the solitary baseline failure the validator trace documents the missing required command sequence and the canonicalized configuration before and after the automatic repair; the post-repair canonicalized configuration satisfied the full intent constraints per the validator. These per-task traces (validator decision fields and repair flags) are the primary provenance for the contingency counts and are archived alongside the analysis artifacts.

Sensitivity and interpretive bounds (post-hoc, artifact-grounded): Because the observed discordant total is one, the maximal absolute improvement observable in this dataset equals 1/300. The paired bootstrap CI [0.00, 0.01] and the exact McNemar p together reflect that any larger effect size is incompatible with the observed discordant evidence. Operationally, this implies that confirmatory designs aiming to test large relative improvements must either (a) calibrate the task bank to produce a higher baseline failure prevalence or (b) adopt independent-draws sampling (increasing discordants at the cost of within-pair control) — choices that should be incorporated into power calculations. These precise numerical bounds are reproduced in the archived paired_contingency_table.csv and paired_difference_figure.svg contained in the analysis bundle.

Reproducibility and archived artifacts: All numerical counts, the bootstrap procedure (seed = 1729, resamples = 10,000), the exact-test implementation, and every per-task validator trace and repair prompt/response are included in the archived execution manifest and analysis bundle. The archived analysis artifacts (condition summary, paired contingency table, bootstrap outputs, and deterministic reconciliation/provider audit) reproduce the figures and numbers above when loaded with the provided analysis executor. Readers who wish to reproduce the exact inferential outputs can consult the execution manifest and the bundled analysis artifacts referenced in the Disclosure Statement.

Concise summary of the results section: The guarded pipeline corrected the single observed invalid baseline output (artifact-grounded repair success = 1/1) but, because the baseline pass rate was far higher than the preregistration planning assumption, the run produced only one discordant pair and therefore lacked statistical information to verify the preregistered >=50% relative reduction. The run nevertheless demonstrates end-to-end execution, archival, and artifact-grounded diagnostics for a generate→validate→repair NetOps pipeline and clarifies concrete sample-size and sampling-semantic constraints necessary for informative confirmatory inference in future studies.

V. Discussion

The guarded pipeline demonstrably remediated the single invalid baseline output observed in this run: the deterministic validator flagged the violation and the single permitted automated repair produced a canonicalized configuration that the validator accepted. This outcome empirically supports the practical utility of a frozen validator plus a tightly bounded repair step as a floor-safety mechanism that can convert some validator-detected failures into valid first-pass configurations.

However, the run also illustrates an important inferential and design caveat: confirmatory power for paired tests depends critically on the number of informative discordant pairs. Our preregistered shared_initial_candidate pairing was chosen to model operational workflows that examine and remediate a single generated plan. That design reduces within-pair heterogeneity and isolates the effect of validation+repair on a given candidate, but it also reduces the expected discordant pair count relative to independent draws per arm. In our execution the baseline pass rate was 299/300, producing only one discordant pair and thereby imposing a strict numerical ceiling

on resolvable absolute improvement (1/300 $\approx$ 0.00333). Consequently, the preregistered >=50% relative-reduction target—formulated under a planning assumption baseline_pass_rate = 0.40—became unattainable given the observed prevalences. The archived preregistration explicitly records the baseline_pass_rate = 0.40 planning assumption and the analysis artifacts show the realized counts; together they explain why the confirmatory hypothesis could not be supported despite the guarded pipeline correcting the single failure it encountered.

Operationally useful lessons flow directly from these artifact-grounded facts. First, when baseline error prevalence is uncertain, researchers should design task banks or sampling procedures to produce an expected number of initial failures (or discordant pairs) aligned with the planned detectable effect. For the preregistered aim in this study, the analysis artifacts indicate that on the order of 90 discordant pairs would have been needed to resolve an absolute change comparable with the preregistered target ($\approx$0.30 absolute). In practice, investigators can (a) increase task difficulty, (b) include adversarial or stress-test augmentations, or (c) adopt stratified sampling by difficulty to concentrate statistical information where it is most informative. These are operational design options rather than claims about guaranteed outcomes; their selection should be preregistered and justified against expected discordant counts.

Second, choice of pairing semantics must be matched to the operational claim. If the goal is to measure whether validator+repair improves the probability that an operator will accept a generated plan when the generator is sampled afresh per condition, independent per-arm draws increase discordance and statistical information for McNemar-style inference. If, instead, the workflow examines and remediates a single proposed plan (the shared_initial_candidate semantics used here), then evaluation budgets should anticipate reduced discordant counts and compensate via deliberately higher-difficulty tasks or larger N. The preregistration tradeoff—higher within-pair control versus greater discordant information—should be made explicit in power computations; our archived power plan assumed baseline_pass_rate = 0.40 and thus did not account for the realized near-ceiling baseline.

Third, repair-effect estimands are inherently conditional on initial failures and require sufficient counts of those failures to estimate repair effectiveness precisely. In this run the observed repair success was 1/1, which is artifact-grounded but statistically imprecise (Clopper–Pearson 95% CI = [0.025, 1.0]). When repair effectiveness is a primary operational claim, studies should be powered to observe an adequate number of initial failures so that conditional repair success has narrow confidence intervals; appropriate sample sizing depends on the target precision and should be preregistered. The deterministic artifacts archived here (per-task validator traces and repair prompts) exemplify the level of per-case traceability required to support conditional estimands and subsequent diagnostics.

Finally, construct and external validity remain constrained in this study. Our validator performed deterministic static canonicalization and rule checks rather than full control-plane or packet-level emulation; therefore, runtime semantics and emergent control-plane interactions were not exercised. The task bank is synthetic and vendor-template coverage was constrained, and only one declared model (gpt-5-mini) and one validator implementation were used. These limitations narrow direct production generalization but do not diminish the methodological point: artifact-grounded preregistration combined with deterministic validator traces yields reproducible counts that clarify what was actually tested and which inferential questions remain open.

In sum, the empirical artifact shows that small guarded pipelines can correct individual validator-detected errors, but proving large relative reductions in misconfiguration rate under paired designs requires aligning task difficulty, pairing semantics, and sample size with the planned estimand. We therefore recommend that future confirmatory NetOps evaluations (a) preregister pairing semantics and the expected discordant-pair count implied by the power calculation, (b) calibrate task difficulty or use adversarial augmentation to achieve a target initial-failure prevalence when necessary, and (c) archive per-task validator and repair traces to support transparent conditional estimands and post-hoc diagnostics. The archived execution and analysis artifacts accompanying this study provide a concrete example of how those recommendations can be operationalized.

VI. Limitations

- Synthetic task bank and constrained vendor-template scope limit external validity to broad production environments.
- Single LLM (gpt-5-mini) and a single validator implementation restrict generality across model families and validator designs.
- The preregistered power plan assumed a baseline pass rate = 0.40; the observed baseline (299/300) contradicts that assumption and removed headroom to detect the preregistered >=50% relative reduction.
- Sampling semantics (shared_initial_candidate) reused the same initial candidate across paired arms, increasing within-pair correlation and reducing discordant pairs available to McNemar inference.
- Validation used deterministic configuration/constraint checks rather than full dynamic emulation of runtime semantic effects; actual runtime consequences of configurations were not executed and remain unmeasured.
- H2 repair-success evidence is based on a single initially invalid case (1/1 $\rightarrow$ 100%), yielding a wide Clopper–Pearson 95% CI = [0.025, 1.0]; larger counts of initial failures are required to estimate repair effectiveness precisely.

VII. Conclusion

We executed a preregistered paired experiment (N=300 pairs) comparing single-shot LLM generation (gpt-5-mini) to a guarded pipeline consisting of a frozen deterministic validator and a single automated repair iteration. Baseline single-shot generation yielded 299/300 valid outputs and the guarded

pipeline yielded 300/300 valid outputs. The single automated repair corrected the lone invalid baseline output, but the observed near-ceiling baseline contradicted the preregistered planning assumption (baseline pass rate = 0.40) and removed statistical headroom to detect the preregistered >=50% relative reduction. Future confirmatory NetOps evaluations should (a) calibrate task difficulty to produce sufficient initial failures or target discordant counts, (b) preregister pairing semantics and repair budgets explicitly, and (c) include emulation to measure runtime semantic effects when operational deployment claims are intended. All raw outputs, validator traces, repair traces, the execution manifest, and analysis artifacts are archived and enumerated in the Disclosure Statement to permit reproduction of the reported counts and intervals.

Disclosure Statement

This study was executed autonomously after the autonomy lock under the archived master prompt (provenance/master_prompt.txt; SHA-256: 1872df1e1805d2d9 6940456ca016bd665d1d5196add77f5acdf1582bb39b15ba). Human role: the Research Director selected the topic family (Generative AI and LLMs for NetOps) and approved the master prompt prior to the autonomy lock; no manual human editing of manuscript technical content, figures, tables, or references occurred after the lock. Preregistration: cand-2026-001-A-prereg-v1 (the preregistration record was created before observing final outcomes; preregistration SHA-256 is recorded in the execution manifest). Declared model and tooling: gpt-5-mini (declared_model_version), adapter family hosted_netops_gvr_v1, frozen deterministic validator/repair adapter deterministic_netops_validator_v5. Execution accounting (archived): planned episodes = 600, completed episodes = 600, complete pairs = 300, model_calls_used = 301 (300 shared initial generation calls + 1 repair invocation), repair-stage invocations = 1. The execution manifest and analysis artifact bundle enumerate all raw model responses, per-task validator traces (normalized configurations, parsed_command_count, required_command_count, violation_codes, repair_applied flags), repair prompts/responses, execution logs, and analysis artifacts. The archived execution manifest and analysis bundle are the authoritative source for the numeric counts reported in this manuscript. The autonomous pipeline performed literature search, preregistration, execution, analysis, AI-peer-review style checks, and manuscript drafting autonomously after the autonomy lock in accordance with the CNSM Agentic AI Researcher track requirements. The preregistered planning assumption used in power calculations (planned baseline pass rate = 0.40) is explicitly reported here because it materially affects interpretation of the confirmatory result.

References

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